Ex. No: 5

Title: Bias–Variance, Overfitting and Underfitting Analysis

Packages: Scikit-learn, NumPy, Matplotlib

AIM:

To study underfitting, good fit and overfitting by fitting polynomial regression models of different degrees on noisy data.

ALGORITHM:

1. Start the program and import NumPy, Scikit-learn and Matplotlib.
2. Create noisy data from a sine curve.
3. Split the data into training and testing sets.
4. Fit polynomial regression models of degree 1, 4 and 15.
5. Find the training and testing Mean Squared Error (MSE) for each model.
6. Plot the three models on the same graph.
7. Display the output and stop.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

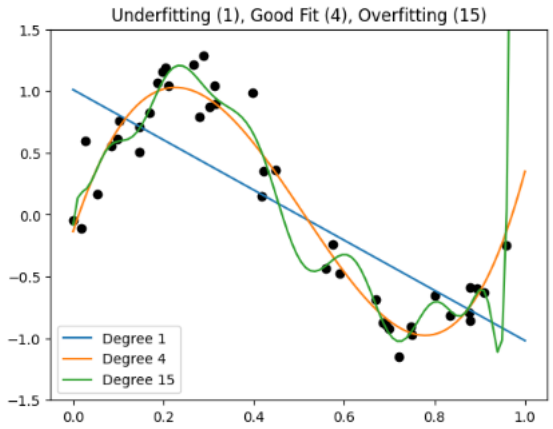
np.random.seed(1)
X = np.sort(np.random.rand(60)).reshape(-1, 1)
y = np.sin(2 * np.pi * X).ravel() + np.random.randn(60) * 0.25
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)

xp = np.linspace(0, 1, 100).reshape(-1, 1)
plt.scatter(X_train, y_train, color="black")
for d in [1, 4, 15]:
    model = make_pipeline(PolynomialFeatures(d), LinearRegression())
    model.fit(X_train, y_train)
    train_err = mean_squared_error(y_train, model.predict(X_train))
    test_err = mean_squared_error(y_test, model.predict(X_test))
    print("Degree", d, "Train MSE =", round(train_err, 3), "Test MSE =", round(test_err, 3))
    plt.plot(xp, model.predict(xp), label="Degree " + str(d))

plt.ylim(-1.5, 1.5)
plt.legend()
plt.title("Underfitting (1), Good Fit (4), Overfitting (15)")
plt.show()
```

OUTPUT:

```
Degree 1 Train MSE = 0.233 Test MSE = 0.358
Degree 4 Train MSE = 0.032 Test MSE = 0.078
Degree 15 Train MSE = 0.022 Test MSE = 34.325
```



RESULT:

Degree 1 showed underfitting (high bias), degree 4 gave a good fit and degree 15 showed overfitting (very low training error but very high testing error).